SPCL-GTM-05 **(FN)**

NARAYANA IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | SPCL-GTM-05 (FN) | Date: 15.01.2025

Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
D	A	B	C	B	B	C	B	A	D
11	12	13	14	15	16	17	18	19	20
B	D	C	D	C	B	B	A	A	A
21	22	23	24	25					
238	4	18	26	18					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
A	D	C	B	C	C	D	A	A	C
36	37	38	39	40	41	42	43	44	45
D	A	B	A	D	A	A	D	C	A
46	47	48	49	50					
5	2	2	150	7					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	C	A	A	D	B	D	B	B	D
61	62	63	64	65	66	67	68	69	70
D	D	D	A	C	D	B	C	B	C
71	72	73	74	75					
3	5	2	6	25					

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PHYSICS

01. $i_L = \frac{15}{10^3} = 15mA.....(1)$

$i_L = \frac{(100-15)}{500} = \frac{85}{5}\times 10^{-2}$

$= 170\,mA$

$i_Z = i - i_L = 170 - 15 = 155\,mA$

02. $from\ I = \frac{dq}{dt} = I_0 - \beta t$

$dq = I_0.dt - \beta t.dt$

$\left[\begin{array}{l} I_0 = 0\ at \\ t = \frac{I_0}{\beta} = \frac{20}{4} = 5\sec \end{array}\right]$

$q = \int_0^5 I_0.dt - \beta\int_0^5 t.dt$

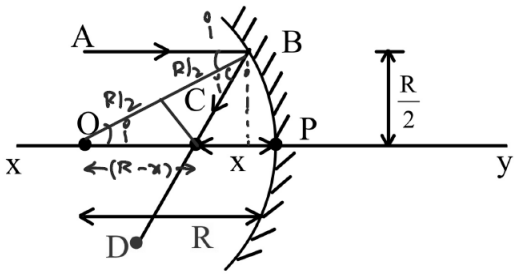
$= 20(5) - 4\left(\frac{5^2}{2}\right)$

$= 100 - 2\times 25$

$= 50C$

03. Tension at a point on rod of (length L) at a distance x from point of application of force is $T = F\left(1-\frac{x}{L}\right)$ both cases. Hence weight has no effect on tension in situation of figure (ii). Extension in rod occurs due to force acting at any point on the rod. in certain cases when net force acts at the centre of rod like weight, extension due to this force may not occur like the given case.

04.



From fig $\cos i = \frac{R/2}{R-x}$

$\sin i = \frac{R/2}{R} = \frac{1}{2} \Rightarrow i = 30^0$

$\therefore \frac{\sqrt{3}}{2} = \frac{R}{2(R-x)} \Rightarrow \sqrt{3}(R-x) = R$

$x = R\left(1-\frac{1}{\sqrt{3}}\right)$

05. The total electric field from Gaussians surface is only due to net charge enclosed with in Gaussians surface.

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06. $v=\sqrt{v_x^2+v_y^2}=\sqrt{(-12\sin 4t)^2+(-12\cos 4t)^2}$

$\Rightarrow v=12\,m/s$

Speed is constant.

Distance=speed$\times$ time $=12\times 2=24m$

07. From x_1 to x_2 interaction force is repulsive, so potential energy decreases and from x_3 to x_4 interaction force is attractive, so kinetic energy decreases.

08. $R=\dfrac{pl}{A}=\dfrac{pl}{\left(\dfrac{\pi d^2}{4}\right)}$

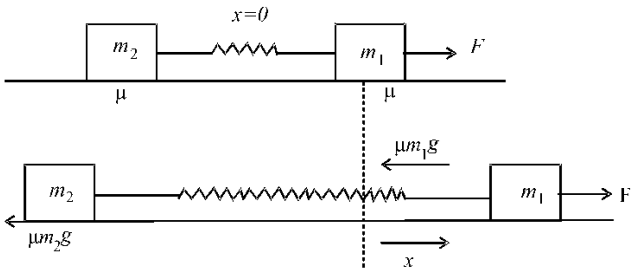
$\dfrac{\Delta\rho}{\rho}\%=\left(\dfrac{\Delta l}{l}\times 100\right)+2.\left(\dfrac{\Delta d}{d}\right)\times 100+\dfrac{\Delta R}{R}\times 100$

$\dfrac{\Delta R}{R}=\dfrac{\Delta V}{V}+\dfrac{\Delta i}{i}$

$\dfrac{\Delta\rho}{\rho}\%=\left(\dfrac{0.1}{10}\times 100\right)+2\times\left(\dfrac{0.1}{2.50}\right)\times 100+\left(\dfrac{0.1}{10}\times 100\right)+\left(\dfrac{0.01}{1.00}\times 100\right)$

09.

$W_F+W_{Sp}+W_{\text{fric}}=\Delta k$



$\Rightarrow Fx-\dfrac{1}{2}kx^2-\mu m_1 gx=0\ and\ kx=\mu m_2 g$

$\Rightarrow F-\dfrac{1}{2}\mu m_2 g-\mu m_1 g=0$

$\Rightarrow F=\mu m_1 g+\dfrac{\mu m_2 g}{2}$

10. The maximum value of electric field is 300 V/m. Therefore the maximum electric force

$F_e=qE_0=1.6\times 10^{-19}\times 300$

$=4.8\times 10^{-17}\,N$

The maximum magnetic field $B_0=\dfrac{E_0}{c}=\dfrac{300}{3\times 10^8}=10^{-8}T$

The maximum magnetic force on electron $F_m=qvB_0$

$=\left(1.6\times 10^{-19}\right)\times\left(2.0\times 10^7\right)\times 10^{-8}$

$=3.2\times 10^{-18}\,N.$

11. Since $a_T=a_N=\dfrac{\upsilon^2}{R}\ and\ a_T=\upsilon\dfrac{d\upsilon}{dS}$

$\Rightarrow \upsilon\dfrac{d\upsilon}{dS}=-\dfrac{\upsilon^2}{R}$

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$$\Rightarrow \int_{\upsilon_0}^{\upsilon} \frac{d\upsilon}{dS} = -\frac{1}{R}\int_0^S dS$$

$$\log_e \upsilon|_{\upsilon_0}^{\upsilon} = -\frac{S}{R}$$

$$\log_e\left(\frac{\upsilon}{\upsilon_0}\right) = -\frac{S}{R}$$

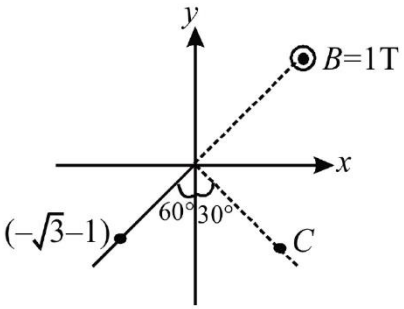
$$\frac{\upsilon}{\upsilon_0} = e^{-\frac{S}{R}}$$

12. As $\vec{M} \| \vec{B}$ $\vec{T}$ is zero

13. $I_g = 10\times10^{-3}/5 = 2\times10^{-3}A;$

$$R = \frac{V}{I_g} - R_g = \frac{1}{2\times10^{-3}} - 5 = 495\Omega.$$

14. The center will be at C as shown:



Coordinates of the center are $\left(r\cos 60^0, -r\sin 60^0\right)$

Where r = radius of circle

15. Power of combinations

$$P = 2(\mu-1)\left(\frac{1}{-60} - \frac{1}{-20}\right) + \left(\frac{-1}{-20/2}\right)$$

$$P = \frac{2}{15} = \frac{-1}{f}$$

$\therefore$ Resultant focal length

$$f = -\frac{15}{2}$$

$\therefore$ From mirror formula

$$\frac{1}{-30} + \frac{1}{v} = \frac{1}{f}$$

Solving $v = -10\,\text{cm}$

Alternatively

Let x be the distance at which object and image coincide, the

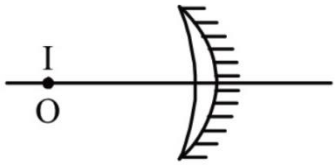
$$\frac{1.5}{-20} - \frac{1}{-x} = \frac{1.5-1}{-60} \Rightarrow x = 15cm$$

$$\therefore f_{eq}(\text{of the mirror} = \frac{15}{2} = 7.5cm$$

Now from the mirror formula

$v = -10cm$

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16. Force on the bullet becomes zero at

$0 = 600 - \left(2 \times 10^5\right) t$

Or $t = 0.003 s$

$\therefore \quad$ impulse $= \int_0^{0.003} F dt$

$= \int_0^{0.003} \left(600 - 2 \times 10^5 t\right) dt = 0.9 N - s$

17. One capacitor is short circuited and the other two are in parallel.

18. We have from the gas law

$\left(P_0 - \alpha V^2\right) V = nRT$

$T = \dfrac{\left(P_0 - \alpha V^2\right) V}{nR} = f(V)$

From the maximum temperature

$\dfrac{dT}{dV} = 0$

$or \;\; P_0 - 3\alpha V^2 = 0$

$V = \left(\dfrac{P_0}{3\alpha}\right)^{1/2}$

Therefore, corresponding pressure

$P = P_0 - \alpha \dfrac{P_0}{3\alpha} = \dfrac{2P_0}{3}$

Hence, the sought maximum temperature from the gas law becomes

$T = \dfrac{PV}{nR} = \dfrac{2\dfrac{P_0}{3}\left(\dfrac{P_0}{3\alpha}\right)^{1/2}}{nR} = \dfrac{2P_0}{3nR}\left(\dfrac{P_0}{3\alpha}\right)^{1/2}$

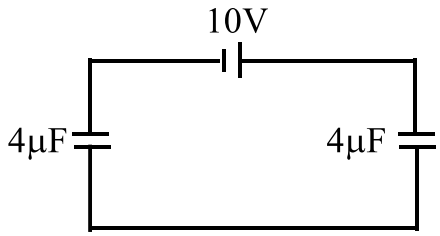
19. $U_g = -\dfrac{GM_1M_2}{R}$

$\Rightarrow U_f - U_i = -\dfrac{GMm}{2R} - \left(-\dfrac{GMm}{2R}\right)$

$\Rightarrow \Delta U = -\dfrac{GMm}{2R} - \dfrac{mgR}{2} \left\{g = \dfrac{GM}{R^2}\right\}$

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20. The given circuit can be simplified as



Now potential difference across each capacitor is 10/2 or 5 V

$\therefore q = CV$

$= (4\mu F)(5V) = 20\mu C$

21. Given ${}_1H^2 + {}_1H^2 \longrightarrow {}_2H^4 + Q$

The total binding energy of the deuterons

$= 4 \times 1.15 = 4.60\,MeV$

The total binding energy of alpha particle

$= 4 \times 7.1 = 28.4\,MeV$

The energy released in the process

$= 28.4 - 4.60 = 23.8\,MeV.$

22. $eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$

$e\frac{V}{3} = \frac{hc}{2\lambda} - \frac{hc}{\lambda_0}$

Solving these two equations we get,

$\lambda_0 = 4\lambda$

23. Horizontal velocity

$v_x = u_x = 18\,m/s$

Now, $\tan 45^0 = \frac{v_y}{v_x}$

$\therefore v_y = v_x = 18\,m/s$

24. $|e_{ind}| = \frac{d\phi}{dt} = \frac{d}{dt}(BA) = A\frac{dB}{dt}$

$= A\frac{d}{dt}(4t^2 + 2t + 5) = A(8t + 2)$

$= 0.06(8t + 2) = 2.52 at\, t = 5$

$i = \frac{2 - e_{ind}}{2} = 0.26A$

25. $E_1 = \frac{1}{2}Kx^2 = 2$

$x = \frac{2}{\sqrt{K}}$

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$$E_2 = \frac{1}{2}Ky^2 = 8 \Rightarrow y = \frac{4}{\sqrt{K}}$$

$$E = \frac{1}{2}K(x+y)^2 = \frac{1}{2}K\left[\frac{2}{\sqrt{K}} + \frac{4}{\sqrt{K}}\right]^2$$

$$E = 18\ J$$

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CHEMISTRY

26. Electron withdrawing group (CI) increases acidic strength while electron donating group $(-CH_3)$ decreases acidic strength. Also magnitude of inductive effect decreases drastically on increasing distance, IV is stronger acid than III.

27. The hydrogen bonds are parallel to the axis of the helix.

28. (i) Terminal alkyne possess acidic hydrogen.
(ii) Iodoform test
(iii) Seliwanoff test
(iv) Picric acid can liberate CO_2 being stronger acid than carbonic acid

29. $\left[CrO_4^{-2}\right]=2\times10^{-4}$

$Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{-2}$

$\left[CrO_4^{-2}\right]=2\times10^{-4}=S$

$K_{sp}=4S^3=4\times\left(2\times10^{-4}\right)^3$

$=4\times\left(2\times10^{-4}\right)^3$

$=4\times8\times10^{-12}$

$=3.2\times10^{-11}$

30. NCERT examples.

31. The IUPAC name of $NaBH_4$ is sodium tetrahydridoborate (III).

32. Since, HCl is not present, so common ion effect does not operate and H_2S ionises to give enough S^{2-} ions. Hence, all Zn^{2+}, Cu^{2+} and Ni^{2+} get precipitated as sulphides.

33. Conceptual

34. conceptual

35. $\Lambda_C=\dfrac{K\times1000}{N}$

$N=\dfrac{K\times1000}{\Lambda_C}$

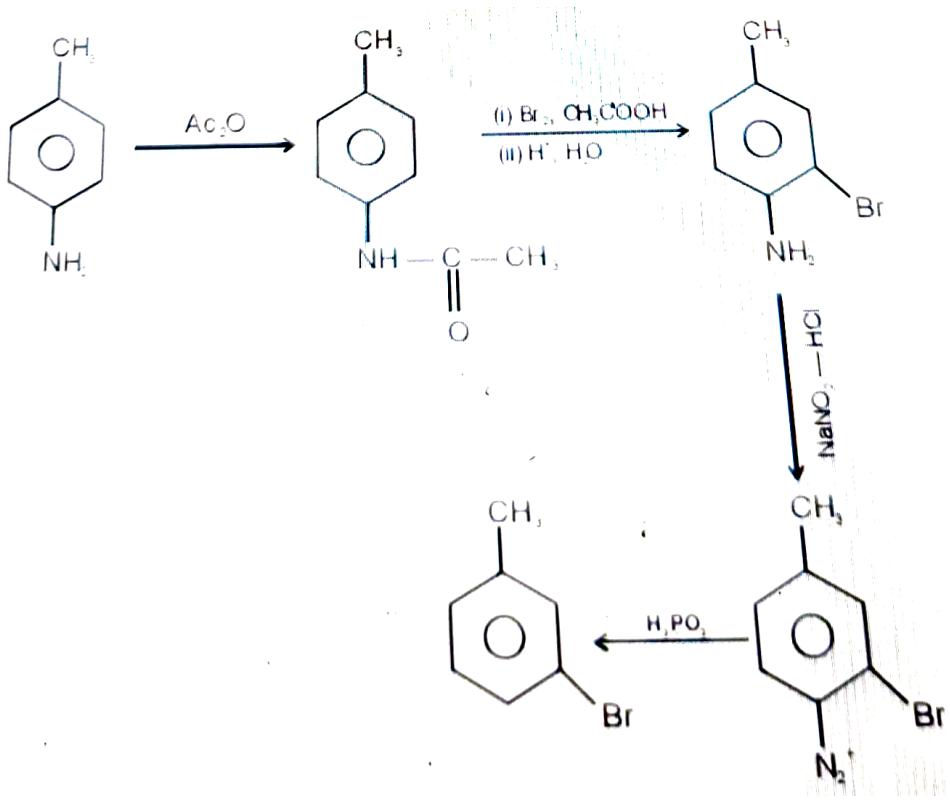
$=\dfrac{3.06\times10^{-6}\times1000}{1.53}=2\times10^{-3}$

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For $BaSO_4$ $M = \frac{N}{2} = 10^{-3}$

$K_{sp} = S^2 = \left(10^{-3}\right) = 10^{-6}$

36.



37. Conceptual

38. VCl_5 do not form

39. Chromatography paper contains water trapped in it, which acts as the stationary phase.

40. Conceptual

41. $K_1 = 10^{-2}$

$K_2 = 1$

$K_3 = 10^4$

Higher equilibrium constant greater is the extent of completion of reaction.

42. Tertiary halide would react at fastest rate with $AgNO_3$ as reaction will proceed by S_N1 mechanism.

43. Conceptual

44. $F(g) + e^- \longrightarrow F^-(g), H_{eg} = -x_1$

$Cl(g) + e^- \longrightarrow Cl^-(g), \Delta H_{eg} = -x_2$

Electron affinity is the energy released $(\Delta H < 0)$ when an electron is added to the gaseous atom to form gaseous anion.

Electron affinity depends on

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(i) Force of attraction, $F = \frac{z_1 z_2 e^2}{r^2}$

(ii) Atomic number (Z).

Smaller the size of the atom, greater the force of attraction and thus, larger the energy released (more negative).

Electron affinity of F should be higher than that of Cl .

But

$\Delta H_{eg}(F) = -328 \text{ kJ mol}^{-1}$

$\Delta H_{eg}(\text{Cl}) = -349 \text{ kJ mol}^{-1}$

It is due to the greater effectiveness of 2 p electrons in small F-atom to repel the additional electron entering the atom than 3 -electrons in the larger Cl -atom.

Thus, Statement l is correct and Statement II is incorrect.

45. Total 5 isomers possible with one exhibiting optical isomerism.

46. NO (15 electrons)

$(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_x)^2(\pi 2p_y)^2(\pi 2p_z)^2(\pi^* 2p_y)^1$

MO without (*) are bonding MO.

47.

$$\mathbf{Cu(s) + 2Ag^+(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)}$$

$$\mathbf{k = 2 \times 10^{15}}$$

$$\mathbf{\frac{1}{2}Cu(s) + Ag^+(aq) \rightleftharpoons Cu^{+2}(aq) + 2Ag(s)}$$

$$\mathbf{K' = \frac{1}{(K)^{1/2}} = \frac{1}{(2 \times 10^{15})^{1/2}}}$$

$$\mathbf{= 2.23 \times 10^{-8}}$$

$$\mathbf{x = 2}$$

48. First order rate equation

49. $q = 0, \Delta U = w$

$1 \times 20 \times [T_2 - 300] = -3000$

$T_2 - 300 = -150 \Rightarrow T_2 = 150K.$

50. (7) $CH_2Cl_2, H_2O, CHCl_3$, O-cresol, SCl_2, IBr and HCHO (in all seven) have dipole moments.

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MATHEMATICS

51. $\left|P^TAP-2I\right|=\left|P^TAP-2P^TP\right|$

$\left|P^T\right|\left(\left|A-2I\right|\right)\left|P\right|$

=27

Sum of divisor $=1+3+9+27=40$

52. ${}^{7+3-1}C_{3-1}={}^{9}C_2=36$

53. D.r's of $\overline{AB}$ are $(1,4,4-\lambda)$ and D.r's of $\overline{BC}$ are $(\lambda-2,2,-2)$

Since $\underline{B}=\frac{\pi}{2}$

$\lambda-2+2(4)-2(4-\lambda)=0\Rightarrow\lambda=\frac{2}{3}$

54. $\sum x_i=15\times12$ and $\frac{\sum x_i^2}{15}-12^2=14$

And $\sum y_i=15\times14$ and $\frac{\sum y_i^2}{15}-14^2=\sigma^2$

Now $13=\frac{(14+144)\times15+\left(\sigma^2+196\right)\times15}{30}-13^2\Rightarrow\sigma^2=10$

55. Drawing the graph a = 3, b = 4

56. $Z=\frac{cis\left(-\frac{2\pi}{5}\right)}{cis\left(\frac{2\pi}{5}\right)}=cis\left(-\frac{4\pi}{5}\right)$

$Z^2=cis\left(-\frac{8\pi}{5}\right)=cis\left(\frac{2\pi}{5}\right)$

$r=1,\theta=\frac{2\pi}{5}$

57. $1=x^3-13x+13$

$\Rightarrow x^3-13x+12=0$

Roots are 1,3, -4

58. Area of quadrilateral ABCD = Area of ΔABC + Area of ΔACD

$=\frac{1}{2}\left[a\times(2a+3b)\right]+\frac{1}{2}\left[(2a+3b)\times b\right]$

$=\frac{1}{2}\left[3a\times b+2a\times b\right]=\frac{5}{2}a\times b$

$=\frac{5}{2}area\,of\,\mathrm{P}_{gram}\ ABCD\ \therefore\ \alpha=\frac{5}{2}$

59. Let d be the common difference of the A.P.

Then, $\log_y x=1+d\Rightarrow x=y^{1+d}$

$\log_z y=1+2d\Rightarrow y=z^{1+2d}$

And $-15\log_x z=1+3d\Rightarrow z=x^{-(1+3d)/15}$

$x=y^{1+d}=z^{(1+2d)(1+d)}$

$=x^{-(1+d)(1+2d)(1+3d)/15}$

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$\Rightarrow (1+d)(1+2d)(1+3d) = -15 \Rightarrow d = -2$

$x = y^{1+d} = y^{-1}, y = z^{1+2d} = z^{-3}$ and $x = \left(z^{-3}\right)^{-1} = z^3$

60. $\int \frac{\cos\theta}{3+\sin(2\theta)}d\theta = \frac{1}{2}\int \frac{\sin\theta+\cos\theta}{4-(\sin\theta-\cos\theta)^2}d\theta + \frac{1}{2}\int \frac{\cos\theta-\sin\theta}{2+(\sin\theta+\cos\theta)^2}d\theta$

$= \frac{1}{8}\log\left(\frac{2+\sin\theta-\cos\theta}{2-\sin\theta+\cos\theta}\right) + \frac{1}{2\sqrt{2}}\tan^{-1}\left(\frac{\sin\theta+\cos\theta}{\sqrt{2}}\right)$

61. $S_{11} > 0$ out side

$S_{11} < 0$ inside

62. $y\cos\left(\frac{y}{x}\right)(xdy - ydx) + x\sin\left(\frac{y}{x}\right)(xdy + ydx) = 0$

$\Rightarrow \frac{dy}{dx} = \frac{\left(\frac{y}{x}\right)^2 \cos\frac{y}{x} - \frac{y}{x}\sin\frac{y}{x}}{\frac{y}{x}\cos\frac{y}{x} + \sin\frac{y}{x}}$

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\Rightarrow v + x\frac{dv}{dx} = \frac{v^2\cos v - v\sin v}{v\cos v + \sin v} \Rightarrow x\frac{dv}{dx} = -\frac{2v\sin v}{v\cos v + \sin v}$

$\Rightarrow \int\left(\frac{v\cos v + \sin v}{v\sin v}\right)dv = -\int\frac{2}{x}dx$

$\Rightarrow \ln\sin v + \ln v = -\ln x^2 + \ln c$

$\Rightarrow (v\sin v)x^2 = c$ given $y(1) = \frac{\pi}{2} \Rightarrow c = \frac{\pi}{2}$

$\Rightarrow (v\sin v)x^2 = \frac{\pi}{2} \Rightarrow xy\sin\frac{y}{x} = \frac{\pi}{2}$

63. ${}^nC_0 + {}^nC_2\frac{1}{2^2} = 2\,{}^nC_1\frac{1}{2}$

$\Rightarrow 1 + \frac{n(n-1)}{8} = n = \frac{n^2-n+8}{8}$

$\Rightarrow n^2 - 9n + 8 - 0 \Rightarrow n = 1, 8 \quad n = 1$ (rejected)

$\therefore$ sum of coefficient $= \left(1+\frac{1}{2}\right)^8 = \left(\frac{3}{2}\right)^8$

64. Lines intersected at points respectively (3,1) and (2,1) eq of common line is $y - 1 = 0$. The distance from (1,2) is 1.

65. $\cos\left(2\sin^{-1}\frac{2}{3}\right) = \frac{1}{9}$, $\cos\left(2\tan^{-1}\frac{1}{3}\right) = \frac{4}{5}$

Q.E is $x^2 - \left(\frac{41}{45}\right)x + \frac{4}{45} = 0$

66. $f'(x) = \frac{\log x - 1}{(\log x)^2}$

$f'(x) = 0 \Rightarrow x = e$

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$f(x)$ is increasing $\Rightarrow f'(x)>0\Rightarrow x>e$

$f(x)$ is decreasing $\Rightarrow f'(x)<0\Rightarrow 0<x<e,\ x\neq 1$

67. $y=\dfrac{\left(x^2-\sqrt{2}x+1\right)\left(x^2+\sqrt{2}x+1\right)}{\left(x^2+\sqrt{2}x+1\right)}$

$y=x^2-\sqrt{2}x+1$

$y^1=2x-\sqrt{2}$

$y^{11}=2$

68. $T=\{(x,y):x-y\in I\}$ as $0\in I,T$ is a reflexive relation.

If $x-y\in I\quad y-x\in I$

T is symmetrical also,

If $x-y=I_1$ and $y-z=I_2$

Then $x-z=(x-y)+(y-z)=I_1+I_2\in I$

T is also transitive

Hence, T is an equivalence relation.

Clearly $x\neq x+1\Rightarrow(x,x)\notin S$

S is not reflexive

69. Required probability

$=P(K\cup H\cup R)=P(K)+P(H)+P(R)-P(K\cap H)-P(K\cap R)-P(H\cap R)+P(K\cap H\cap R)$

$=\dfrac{4}{52}+\dfrac{13}{52}+\dfrac{26}{52}-\dfrac{1}{52}-\dfrac{2}{52}-\dfrac{13}{52}+\dfrac{1}{52}=\dfrac{7}{13}$

70. $\vec{r}=\vec{d}=d_1\hat{i}+d_2\hat{j}+d_3\hat{k}$

$d_1^2+d_2^2+d_3^2=51$

Each of the given vectors a, b and c is a unit vectors

$\cos\theta=\dfrac{\vec{d}.\vec{a}}{|d||a|}=\dfrac{\vec{d}.\vec{b}}{|d||b|}=\dfrac{\vec{d}.\vec{c}}{|d||c|}$ or $\vec{d}.\vec{a}=\vec{d}.\vec{b}=\vec{d}.\vec{c}$

$|d|=\sqrt{51}$ Cancel out and $|a|=|b|=|c|=1$

Hence $\dfrac{1}{3}(d_1-2d_2+2d_3)=\dfrac{1}{5}(-4d_1+0d_2-3d_3)=d_2$

On solving we get $\dfrac{d_1}{5}=\dfrac{d_2}{-1}=\dfrac{d_3}{-5}=\lambda$

Putting d_1,d_2 and d_3 in Eq then $\lambda=\pm 1$

Hence then required vector is $\bar{r}=-5\hat{i}+\hat{j}+5\hat{k}$

71. The given circle is $(x-1)^2+(y-1)^2=1$

Length of the chord $=2\sqrt{r^2-d^2}$

$\Rightarrow$ locus of the point M is $(x-1)^2+(y-1)^2=\dfrac{3}{4}$

72. Solution of system of equation is $(a,b,c)=(\lambda,\lambda-1,\lambda)$

Now $ax+by+c=2$

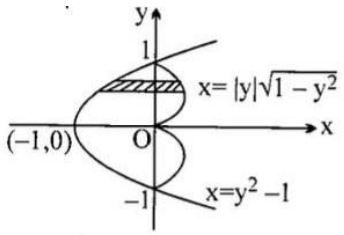
$\Rightarrow\lambda(x+y+1)=(y+2)=0$

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$x=1, y=-2$

$(p,q)=(1,-2)$

73.



$$A = 2\int_0^1 \left[y\sqrt{1-y^2} - (y^2-1) \right] dy$$

$= 2$

74. $\int_0^x ty(t)\,dt = x^2 y(x)$

Diff w. r. t x

$xy = x^2 \frac{dy}{dx} + y(2x)$

$x^2 \frac{dy}{dx} + xy = 0$

$\Rightarrow \frac{dy}{y} + \frac{dx}{x} = 0$

$\Rightarrow xy = c$

It is passing through (2,3) C =6

75. Slope of $3x+2y=1$ is $-\frac{3}{2}$

$P = \lim_{x\to 0} \frac{x(x-1)}{f\left(\frac{e^{2x}}{2}\right) - f\left(\frac{-2x}{2}\right)} \left(\frac{0}{0}\right)$

$\Rightarrow \frac{-1}{f'\left(\frac{1}{2}\right) + f'\left(\frac{1}{2}\right)} = \frac{1}{3}$

$50\sum_{r=1}^{\infty} p^r = 50\frac{1/3}{\left(1-\frac{1}{3}\right)} = 25$

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